# Modelling With Differential Equations

## Chapter Overview

Solution 2

This chapter applies first- and second-order differential equations in practical settings.

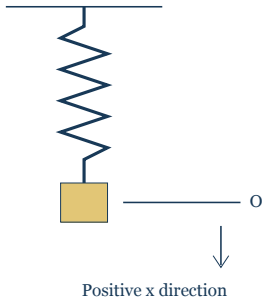
1. First-order modelling

A particle has acceleration $a = t + 3v/t$, $t > 0$, and $v = 0$ when $t = 2$. Show $v = ct^3 - t^2$, finding $c$.

2. Simple harmonic motion

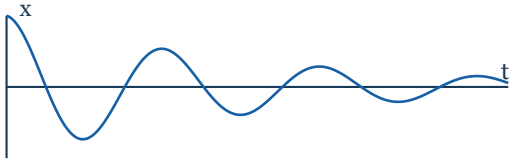
Acceleration towards a central point is proportional to displacement:

$\ddot{x} = -\omega^2 x$.




3. Damped and forced harmonic motion

Model damping forces and additional external forces.




4. Coupled first-order equations

Differential-equation versions of simultaneous equations, including population models:

$\dot{x} = ax + by, \quad \dot{y} = cx + dy$.

## First-order Models

Solution 3

Pure Year 2 introduced forming and solving first-order models. Here these may have the form $y' + Py = Q$. Motion gives examples such as $a = dv/dt$.

A particle starts from rest at $O$, moving along a line with acceleration $6/(t+2)^2$ m s$^{-2}$, $t \geq 0$.

(a) Find its velocity.

(b) Show its displacement at $t = 6$ is $18 - 12\ln 2$ m.

1a

$$\frac{dv}{dt} = \frac{6}{(t+2)^2} \Rightarrow v = -\frac{6}{t+2} + c.$$

$$v(0) = 0 \Rightarrow c = 3 \Rightarrow v = 3 - \frac{6}{t+2}.$$

1b

$$\frac{ds}{dt} = 3 - \frac{6}{t+2}$$

$$s = 3t - 6\ln(t+2) + 6\ln 2$$

$$s(6) = 18 - 6\ln 8 + 6\ln 2 = 18 - 12\ln 2.$$

A particle has acceleration $a = t + 3v/t$, $t > 0$. Given $v(2) = 0$, show $v = ct^3 - t^2$, finding $c$.

Hint: write in the form $dv/dt + Pv = Q$ and multiply by $e^{\int P\,dt}$.

2

$$\frac{dv}{dt} - \frac{3}{t}v = t$$

$$\text{IF} = e^{\int -3/t\,dt} = e^{-3\ln t} = t^{-3}$$

$$\frac{d}{dt}(t^{-3}v) = t^{-2}$$

$$t^{-3}v = -t^{-1} + c.$$

$$v(2) = 0 \Rightarrow c = \tfrac{1}{2}$$

$$v = \tfrac{1}{2}t^3 - t^2.$$

## Filling a Container

Solution 4

A tank initially contains 1000 litres of pure water. Liquid leaves at 30 litres/hour and solution enters at 40 litres/hour, containing 4 grams of copper sulphate per litre. It mixes immediately throughout the tank. Let $x$ be the copper sulphate mass in grams after $t$ hours.

(a) Show $dx/dt = 160 - 3x/(100+t)$.

(b) Find the mass after 6 hours.

(c) Explain how the model could be refined.

a

Volume: $1000 + 40t - 30t = 1000 + 10t$ litres.

Concentration: $x/(1000+10t)$ g/litre.

Inflow: $40 \times 4 = 160$ g/hour.

Outflow: $30x/(1000+10t) = 3x/(100+t)$ g/hour.

$$\frac{dx}{dt} = 160 - \frac{3x}{100+t}.$$

b

$$\frac{dx}{dt} + \frac{3}{100+t}x = 160$$

$$\text{IF} = e^{\int 3/(100+t)\,dt} = (100+t)^3$$

$$(100+t)^3x' + 3(100+t)^2x = 160(100+t)^3$$

$$\frac{d}{dt}\{(100+t)^3x\} = 160(100+t)^3$$

$$(100+t)^3x = 40(100+t)^4 + c.$$

$$x(0) = 0 \Rightarrow c = -4 \times 10^9$$

$$106^3x = 40(106)^4 - 4 \times 10^9$$

$$x \approx 882 \text{ g (3 s.f.)}.$$

c

Allow for the time taken for the copper sulphate to disperse, instead of assuming instantaneous mixing.

Exercise 8A — Core Pure Book 2, pages 174–175.



Exercise 8A — Core Pure Book 2, pages 174–175.

Exercise 8A — Core Pure Book 2, pages 174–175.

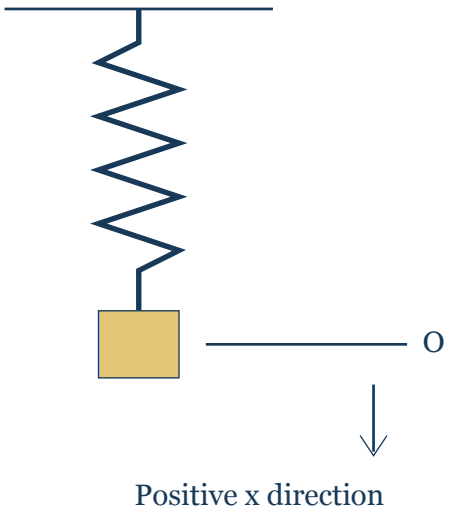


Exercise 8A — Core Pure Book 2, pages 174–175.

## Exercise 8A — Core Pure Book 2 Pages 174–175 Solution 9

Exercise 8A — Core Pure Book 2, pages 174–175.

## Simple Harmonic Motion Solution 10




$O$ is the centre of oscillation. Moving away from $O$, the particle slows down because its acceleration is towards $O$. Doubling displacement doubles acceleration magnitude. It is not constant as in uniform gravity.

Simple harmonic motion (SHM) has acceleration towards a fixed point on the line of motion, proportional to displacement from that point:

$\ddot{x} = -\omega^2 x.$

Dots mean time derivatives; $\ddot{x}$ is acceleration. The minus sign makes it restoring. $\omega > 0$ is angular frequency in radians/second, and $\omega^2$ the proportionality constant.

The chain rule gives

$$\ddot{x} = \frac{dv}{dt} = \frac{dv}{dx}\frac{dx}{dt} = v\frac{dv}{dx}.$$

This helps solve the motion equation.

## Solving the SHM Equation

Solution 11

A particle has SHM about $O$, with maximum displacement $a$.

(a) Show $v^2 = \omega^2(a^2 - x^2)$.

a

$\ddot{x} = -\omega^2 x \Rightarrow v\dfrac{dv}{dx} = -\omega^2 x$

$\displaystyle\int v\,dv = -\int \omega^2 x\,dx$

$\frac{1}{2}v^2 = -\frac{1}{2}\omega^2 x^2 + c.$

At maximum displacement $x = a$, $v = 0$, so $c = \frac{1}{2}\omega^2 a^2$.

$\frac{1}{2}v^2 = -\frac{1}{2}\omega^2 x^2 + \frac{1}{2}\omega^2 a^2$

$v^2 = \omega^2(a^2 - x^2).$

Complex numbers provide a practical tool for solving this applied motion equation.

(b) Show $x = a\sin(\omega t + \alpha)$, with arbitrary phase $\alpha$.

b

$\ddot{x} + \omega^2 x = 0, \quad m^2 + \omega^2 = 0$

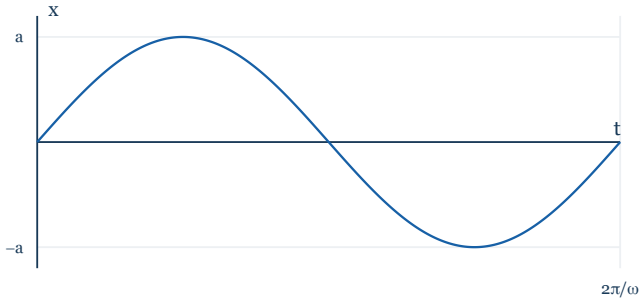
$m = \pm i\omega \Rightarrow x = A\cos\omega t + B\sin\omega t$

$= R\sin(\omega t + \alpha).$

Using Pure Year 2 trigonometry, $R = \sqrt{A^2 + B^2}$.

Since the amplitude is $a$,

$x = a\sin(\omega t + \alpha).$




Displacement follows a sine wave. Amplitude is maximum displacement. The phase $\alpha$ controls translation; $\alpha = 0$ starts at equilibrium. Period $T = 2\pi/\omega$; $\omega = 1$ gives one oscillation per $2\pi$ seconds.

## SHM Examples

Solution 12

A particle moves along a line with $\ddot{x} = -4x$. At $t = 0$, $x = 1$ m and $v = 4$ m s$^{-1}$.

(a) Find displacement after $t$ seconds.

(b) Find maximum displacement from $O$.

Hint: find constants in the sine/cosine form before combining.

1a

$\ddot{x} + 4x = 0 \Rightarrow m^2 + 4 = 0 \Rightarrow m = \pm 2i$

$x = A\cos 2t + B\sin 2t.$

$x(0) = 1 \Rightarrow A = 1$

$v = -2\sin 2t + 2B\cos 2t$

$v(0) = 4 \Rightarrow B = 2$

$x = \cos 2t + 2\sin 2t.$

1b

$x = \sqrt{5}\sin(2t + 0.4636\ldots).$

Maximum displacement is $\sqrt{5}$ m.

A particle joins two identical elastic springs fixed at $A, B$. It is held at a point $C$ between them, with $AC \neq BC$, and released from rest. Let $x$ be displacement from equilibrium $O$, with release point $C$ at $x = 0.4$ m. The model is $\ddot{x} = -25x$.

(a) Describe the motion.

(b) Given $x(0) = 0.4, v(0) = 0$, solve for $x(t)$.

(c) State the period and maximum speed.

2a Simple harmonic motion.

2b

$m^2 + 25 = 0 \Rightarrow m = \pm 5i$

$x = A\cos 5t + B\sin 5t, \quad A = 0.4$

$v = -2\sin 5t + 5B\cos 5t, \quad B = 0$

$x = 0.4\cos 5t.$

2c

$T = \dfrac{2\pi}{5}$ s, $\quad v = -2\sin 5t.$

Maximum speed is 2 m s$^{-1}$.

Exercise 8B — Core Pure Book 2, pages 178–180.



Exercise 8B — Core Pure Book 2, pages 178–180.

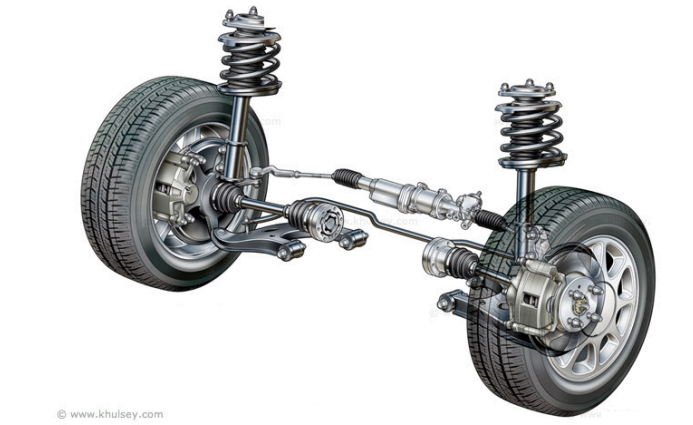
Exercise 8B — Core Pure Book 2, pages 178–180.



Exercise 8B — Core Pure Book 2, pages 178–180.

## Damped Harmonic Motion

Solution 17




In practice the amplitude often decreases over time, so there is another force: damping, opposing velocity and proportional to its magnitude.

The spring's elastic force is called the restoring force.

Without damping, spring extension/compression gives a restoring acceleration proportional to displacement, resulting in sinusoidal motion.

With damping:

$\ddot{x} = -k\dot{x} - \omega^2 x$

$\ddot{x} + k\dot{x} + \omega^2 x = 0, \quad k > 0.$

The roots of the auxiliary equation determine the type of damped motion.

## Types of Damping

Solution 18

$\ddot{x} + k\dot{x} + \omega^2 x = 0, \quad k, \omega > 0.$

$m^2 + km + \omega^2 = 0.$

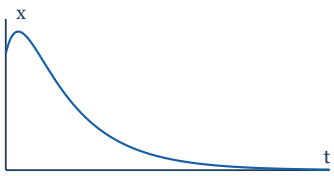
The discriminant determines the nature of the roots.

Distinct real roots:

$k^2 - 4\omega^2 > 0$

$x = Ae^{-\alpha t} + Be^{-\beta t}, \quad \alpha, \beta > 0.$

Heavy damping: no repeated oscillations.

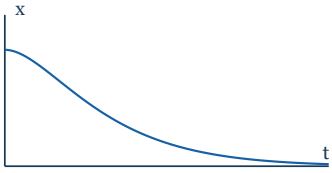



Equal real roots:

$k^2 - 4\omega^2 = 0$

$x = (A + Bt)e^{-kt/2}.$

Critical damping: the boundary between oscillatory and non-oscillatory motion.



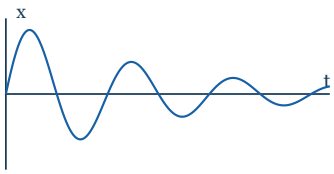

Complex conjugate roots:

$k^2 - 4\omega^2 < 0$

$x = e^{-kt/2}(A\cos \Omega t + B \sin \Omega t)$

$\Omega = \sqrt{\omega^2 - k^2/4}.$

Light damping: oscillations with decreasing amplitude.



## Deriving the Model from Forces

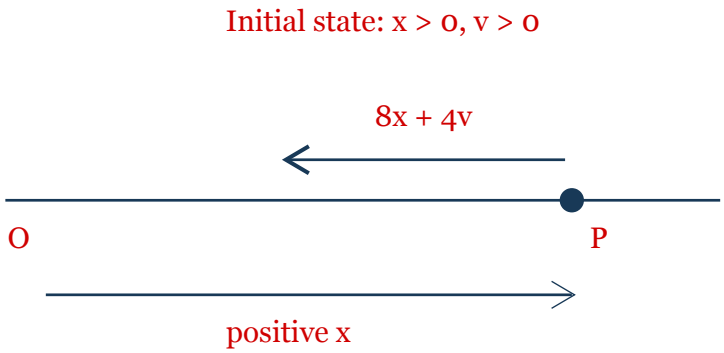
Solution 19

A particle of mass 0.5 kg moves horizontally. Its displacement from $O$ is $x$ m and velocity $v$ m s$^{-1}$. A restoring force has signed value $-8x$ N and resistance $-4v$ N. Initially $x = 1.5$ and it moves in increasing $x$ with speed 4 m s$^{-1}$.

(a) Show $\ddot{x} + 8\dot{x} + 16x = 0$.

(b) Find $x(1)$.

Force diagram




a

$F = ma: \quad -(8x + 4v) = 0.5a$

$0.5a + 4v + 8x = 0$

$\ddot{x} + 8\dot{x} + 16x = 0.$

b

$m^2 + 8m + 16 = 0 \Rightarrow m = -4$

$x = (A + Bt)e^{-4t}.$

$x(0) = 1.5 \Rightarrow A = 1.5$

$v = Be^{-4t} - 4(A + Bt)e^{-4t}$

$v(0) = 4 \Rightarrow B = 10$

$x = (1.5 + 10t)e^{-4t}$

$x(1) = 11.5e^{-4} \approx 0.211$ m (3 s.f.).

Equal roots give critical damping.

## Further Damping Examples

Solution 20

A particle hangs in equilibrium on a light elastic string fixed at $A$. It is pulled down into a resisting liquid and released from rest. While the string is taut and the particle remains submerged,

$\ddot{x} + 6k\dot{x} + 5k^2x = 0, \quad k > 0.$

Find the general solution and type of damping.

$m^2 + 6km + 5k^2 = (m + 5k)(m + k) = 0$

$m = -5k, -k$

$x = Ae^{-5kt} + Be^{-kt}.$

Two distinct real roots give heavy damping.

A particle hangs in equilibrium from a light elastic spring fixed at $A$. It is pulled down and released from rest, with resistance proportional to speed. Its displacement from equilibrium obeys

$\ddot{x} + 2k\dot{x} + 2k^2x = 0, \quad k > 0.$

Find the general solution and period of oscillation.

$m^2 + 2km + 2k^2 = 0$

$m = \dfrac{-2k \pm \sqrt{4k^2 - 8k^2}}{2} = -k \pm ik$

$x = e^{-kt}(A\cos kt + B\sin kt).$

$T = \dfrac{2\pi}{k}.$

The coefficient of $t$ in the sine/cosine gives the angular frequency of the oscillation.

## Forced Harmonic Motion

Solution 21

An additional external force produces forced harmonic motion:

$\ddot{x} + k\dot{x} + \omega^2 x = f(t)$.

This is non-homogeneous, so use CF + PI.

A particle of mass 1.5 kg moves on the $x$-axis under restoring force $-7.5x$ N, resistance $-6\dot{x}$ N and external force $12\sin t$ N in the positive direction. Initially $x = 5, \dot{x} = 2$.

(a) Show $\ddot{x} + 4\dot{x} + 5x = 8\sin t$.

(b) Find $x(t)$.

(c) Describe the motion for large $t$.

a

$-7.5x - 6\dot{x} + 12\sin t = 1.5\ddot{x}$

$\ddot{x} + 5x + 4\dot{x} = 8\sin t$

$\ddot{x} + 4\dot{x} + 5x = 8\sin t.$

b

$m^2 + 4m + 5 = 0 \Rightarrow m = -2 \pm i$

$x_c = e^{-2t}(A\cos t + B\sin t).$

Try $x_p = p\sin t + q\cos t$:

$\dot{x}_p = p\cos t - q\sin t, \quad \ddot{x}_p = -p\sin t - q\cos t.$

$(-p\sin t - q\cos t) + 4(p\cos t - q\sin t)$

$+5(p\sin t + q\cos t) = 8\sin t.$

$p + q = 0, \quad p - q = 2 \Rightarrow p = 1, q = -1.$

$x = e^{-2t}(A\cos t + B\sin t) + \sin t - \cos t.$

$x(0) = 5 \Rightarrow A - 1 = 5 \Rightarrow A = 6$

$\dot{x}(0) = 2 \Rightarrow -2A + B + 1 = 2 \Rightarrow B = 13$

$x = e^{-2t}(6\cos t + 13\sin t) + \sin t - \cos t.$

c

As $t \to \infty$, $e^{-2t} \to 0$. Hence the transient decays and

$x \sim \sin t - \cos t = \sqrt{2}\sin(t - \pi/4)$.

The limiting motion is SHM with amplitude $\sqrt{2}$ m and period $2\pi$ s.

## A Further Forced-motion Model

Solution 22

A particle $P$ is attached to end $A$ of a light elastic string $AB$, initially at rest on a smooth horizontal plane. At $t = 0$, end $B$ moves at constant speed $U$ in direction $AB$. Air resistance is proportional to the particle's speed. For the prescribed model, displacement $x$ from the initial position obeys

$\ddot{x} + 2k\dot{x} + k^2 x = 2kU, \quad k > 0.$

Find $x(t)$, using initial rest: $x(0) = 0, \dot{x}(0) = 0$.

Solution

$m^2 + 2km + k^2 = (m + k)^2 = 0$

$\text{CF} = (A + Bt)e^{-kt}.$

Since the forcing is constant, try $x_p = c$:

$\dot{x}_p = \ddot{x}_p = 0, \quad k^2 c = 2kU \Rightarrow c = \dfrac{2U}{k}.$

$x = (A + Bt)e^{-kt} + \dfrac{2U}{k}.$

$x(0) = 0 \Rightarrow A = -\dfrac{2U}{k}$

$\dot{x} = -k(A + Bt)e^{-kt} + Be^{-kt}$

$\dot{x}(0) = 0 \Rightarrow 0 = -kA + B \Rightarrow B = -2U$

$x = \left(-\dfrac{2U}{k} - 2Ut\right)e^{-kt} + \dfrac{2U}{k}.$

Exercise 8C — Core Pure Book 2, pages 184–186.



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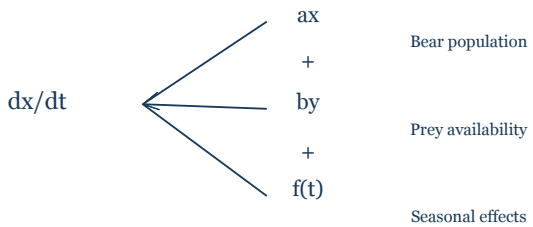
Exercise 8C — Core Pure Book 2, pages 184–186.

## Exercise 8C — Core Pure Book 2 Pages 184–186 Solution 27

Exercise 8C — Core Pure Book 2, pages 184–186.

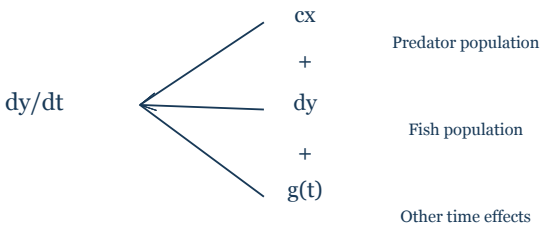
## Coupled First-order Linear Equations Solution 28

Coupled population models describe interacting species. Let $x$ be bears and $y$ fish. Bear population growth depends on bear numbers, prey availability and possibly seasonal effects such as increased breeding.




Fish growth depends on predator numbers (more bears may mean faster decline), fish numbers and other time-dependent factors.

We study a linear model. Classical Lotka–Volterra predator–prey equations are nonlinear; this linear system is a simplified model.



## Solving a Coupled System

Solution 29

$\dot{x} = ax + by + f(t), \quad \dot{y} = cx + dy + g(t).$

Homogeneous if $f = g = 0$ for all $t$.

Strategy: 1. Make $y$ the subject of the first equation and differentiate. 2. Substitute in the second equation and solve the resulting second-order equation for $x$. 3. Differentiate the result and use the first rearrangement to find $y$.

A survey of a remote Canadian island began in 2010. After $t$ years, bear and fish populations $x, y$ obey

$\dot{x} = 0.3x + 0.1y, \quad \dot{y} = -0.1x + 0.5y.$

(a) Show $\ddot{x} - 0.8\dot{x} + 0.16x = 0$.

(b) Find the general solution for bears.

(c) Find the general solution for fish.

(d) Initially 5 bears and 20 fish: predict bears in 2020.

(e) Comment on suitability.

a

$y = 10\dot{x} - 3x \Rightarrow \dot{y} = 10\ddot{x} - 3\dot{x}$

$10\ddot{x} - 3\dot{x} = -0.1x + 0.5(10\dot{x} - 3x)$

$10\ddot{x} - 8\dot{x} + 1.6x = 0$

$\ddot{x} - 0.8\dot{x} + 0.16x = 0.$

b

$m^2 - 0.8m + 0.16 = 0 \Rightarrow m = 0.4$

$x = (A + Bt)e^{0.4t}.$

c

$\dot{x} = 0.4Ae^{0.4t} + 0.4Bte^{0.4t} + Be^{0.4t}$

$y = 10\dot{x} - 3x$

$= 10(0.4A + 0.4Bt + B)e^{0.4t} - 3(A + Bt)e^{0.4t}$

$y = (A + 10B + Bt)e^{0.4t}.$

d

$x(0) = 5 \Rightarrow A = 5$

$y(0) = 20 \Rightarrow 5 + 10B = 20 \Rightarrow B = 1.5$

$x(10) = (5 + 15)e^4 \approx 1092$ bears.

e Both populations grow without limit, so this is unlikely to remain realistic.

## Test Your Understanding

Solution 30

Two barrels contain contaminated water. At time $t$ seconds, contaminant amounts in $A, B$ are $x, y$ ml. Contaminant enters $A$ at 5 ml/s; water moves both ways through hoses and also drains from $A$ out of the system. The model is

$\dot{x} = 5 + \frac{4}{9}y - \frac{1}{7}x \qquad (1)$

$\dot{y} = \frac{3}{70}x - \frac{4}{9}y \qquad (2).$

Show $630\ddot{y} + 370\dot{y} + 28y = 135$.

Hint: make $x$ the subject in (2), differentiate and substitute into (1).

Solution

$\dot{y} + \frac{4}{9}y = \frac{3}{70}x$

$x = \frac{70}{3}\dot{y} + \frac{280}{27}y$

$\dot{x} = \frac{70}{3}\ddot{y} + \frac{280}{27}\dot{y}.$

Substitute into (1):

$\frac{70}{3}\ddot{y} + \frac{280}{27}\dot{y} = 5 + \frac{4}{9}y - \frac{1}{7}(\frac{70}{3}\dot{y} + \frac{280}{27}y)$

$70\ddot{y} + \frac{280}{9}\dot{y} = 15 + \frac{4}{3}y - 10\dot{y} - \frac{40}{9}y$

$630\ddot{y} + 280\dot{y} = 135 + 12y - 90\dot{y} - 40y$

$630\ddot{y} + 370\dot{y} + 28y = 135.$

Exercise 8D — Core Pure Book 2, pages 188–190.



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